DEMO · ANATOMY

Nerves of the arm Understanding the connections

Which nerve supplies which muscle – and **why**. A short demo of what the study-notes skill produces.

- Core rules
- Overview
- Mnemonic
- Arm
- Why box

Contents

0. The core rules

1. Musculocutaneous nerve

2. Radial nerve

3. Overview

This is a short demo. Real notes cover the whole material – here you see the structure: core rules first, then one section per unit with a fact table, why boxes and mnemonics, and an overview at the end.

0. The core rules

Rule 1 – One compartment = one nerve. Muscles in the same fascial compartment usually do the same job and are supplied by the same nerve. → Anterior arm = musculocutaneous nerve, posterior arm = radial nerve.

Rule 2 – Dorsal and ventral layer. In the embryo, every limb bud has a **dorsal muscle layer (extensors)** and a **ventral muscle layer (flexors)**, each with its own nerves:

	Dorsal layer (extensors)	Ventral layer (flexors)
Arm	axillary n., radial n.	musculocutaneous n., median n., ulnar n.

Rule 3 – Innervation shows origin, not function. If a muscle's nerve doesn't fit its function, the muscle comes from the other layer.

Remember: "Tell me your nerve, and I'll tell you where you come from."

1. Musculocutaneous nerve

Muscle	Function
Biceps brachii	Flexion at the elbow, supination, anteversion
Brachialis (lateral part inconstantly also radial n.)	Flexion at the elbow
Coracobrachialis	Anteversion

Why? This is rule 1 in its purest form: the **anterior compartment of the arm** belongs to the musculocutaneous nerve. It pierces the coracobrachialis, runs between biceps and brachialis and supplies all three. Then it becomes a skin nerve of the forearm – hence "musculo-**cutaneous**": first muscle, then skin.

Why can the brachialis get the radial nerve too? Its lateral part borders the **extensor compartment**, where the radial nerve runs. A muscle at the border of two compartments can be supplied by both.

Mnemonic: "**BBC**": **B**iceps, **B**rachialis, **C**oracobrachialis = musculocutaneous nerve.

2. Radial nerve

Muscle	Function
Triceps brachii (all heads)	Extension at the elbow
Brachioradialis	Flexion at the elbow (!)
Extensors of the forearm	Extension of wrist and fingers

Why? The radial nerve is **the** nerve of the dorsal layer of the whole arm: everything at the back that extends.

Why the brachioradialis, although it flexes? It belongs to the **dorsal layer** by development and only became a flexor because it lies in front of the elbow axis. The innervation shows its origin (rule 3), not its function.

Mnemonic: "**Radial = rear**": everything at the back of the arm that extends belongs to the radial nerve.

3. Overview

Function	Muscle group	Nerve	Layer
Elbow flexors	Biceps, brachialis	Musculocutaneous n.	ventral
Elbow extensors	Triceps	Radial n.	dorsal
Exception	Brachioradialis (flexes)	Radial n.	dorsal

> **If you take only one thing away:** the **compartment** decides the nerve. Where the innervation breaks the pattern, the muscle comes from another layer.